

University of Colorado Boulder

3 Courses

Probability Foundations for Data Science and AI

Discrete-Time Markov Chains and Monte Carlo Methods

Statistical Estimation for Data Science and AI

Aug 20, 2026

Olzhas Azirali

has successfully completed the online Specialization

Foundations of Probability and Statistics

This Specialization equips learners with the probability and statistics skills essential for data science and AI. Learners progress from foundational probability concepts to advanced topics including Markov chains, Monte Carlo methods, and statistical estimation. Graduates can quantify uncertainty, analyze dynamic systems, simulate data from complex distributions, and construct and interpret statistical estimators, building the expertise needed for analytical and AI-driven problem solving.

Senior Instructor
University of Colorado Teaching Professor
Associate Department Chair
Undergraduate Studies Chair

Jem Corcoran
Associate Professor
Applied Mathematics

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